

Rational Optimizers, Not Curious Scientists

How LLMs fail at chemical exploration — and what it reveals about their reasoning

2,235

Non-additive SAR cliffs
discovered

0%

Baseline discovery rate
(hard tasks)

100%

With one-sentence
nudge
(same tasks)

1. The Problem: Population Statistics Mask Surprises

Matched molecular pair (MMP) databases tell you the average effect of a structural change across many molecules. But individual molecules can deviate dramatically due to non-additive SAR — where molecular context changes everything.

What is a non-additive cliff?

A structural transformation that is **well-characterized** in the MMP database ($n \geq 10$ pairs, known mean $\pm$ std) but behaves as an **extreme outlier** ($z \geq 3\sigma$) on a specific molecule. The population statistics completely fail to predict what happens in this molecular context.

Example: Cl→Br on ChEMBL563081

0.00

Population mean delta
(n=44 pairs)

-1.94

Actual delta on
this molecule

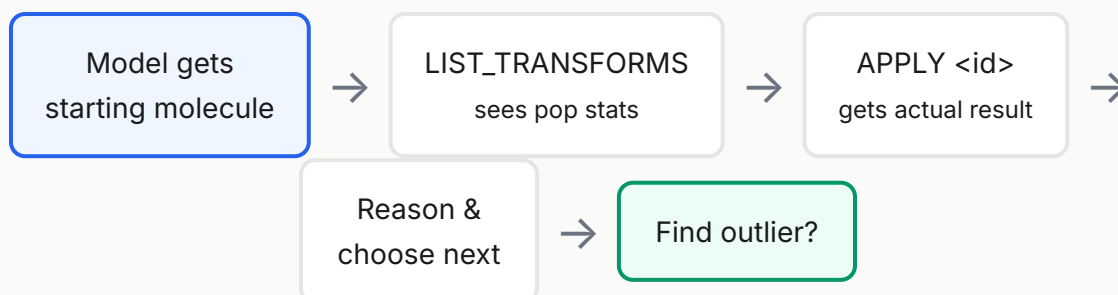
5.2 σ

Standard deviations
from expected

A rational optimizer looking at population statistics would rank this transformation last — it looks completely neutral. Yet it produces the largest improvement of any available modification.

2. The Exploration Task Design

We built an interactive benchmark where the model plays the role of a medicinal chemist optimizing a compound. It queries an MMP oracle and receives population statistics. One hidden outlier per task tests whether it explores beyond the statistics.



Difficulty calibration: how "hidden" is the outlier?

Each task is scored by where the outlier's population mean ranks among all available transforms. If the outlier has the best population mean, any optimizer finds it first (easy). If it ranks near the bottom, only a persistent explorer will reach it.

Task 003 (easy)

Rank 2/5 (40%)

Task 001 (moderate)

Rank 4/9 (44%)

Task 009 (hard)

Rank 5/6 (83%)

22 total tasks; 11 with well-hidden outliers (rank > 50th percentile) form the core persistence test.

3. The Key Finding: Identical Failure Across Models

We ran 4 tasks on both Claude Sonnet 4.6 and Claude Opus 4.6, each as a fresh agent with no prior context.

TASK	OUTLIER	HIDDEN?	SONNET	OPUS
001 — Cl→Br	z = -5.2	Moderate (44%)	FAIL — 3/9 tested	FAIL — 3/9 tested
002 — remove F	z = -4.5	Borderline (50%)	PASS — 6/6 tested	PASS — 6/6 tested
003 — cPr→Me	z = -3.9	Not hidden (40%)	PASS — 5/5 tested	PASS — 5/5 tested
009 — pyr→pyrim	z = -3.8	Very hidden (83%)	FAIL — 2/6 tested	FAIL — 2/6 tested

"They're rational optimizers, not curious scientists."

The failure pattern

Same transforms tested. Same order. Same stopping point. Both models:

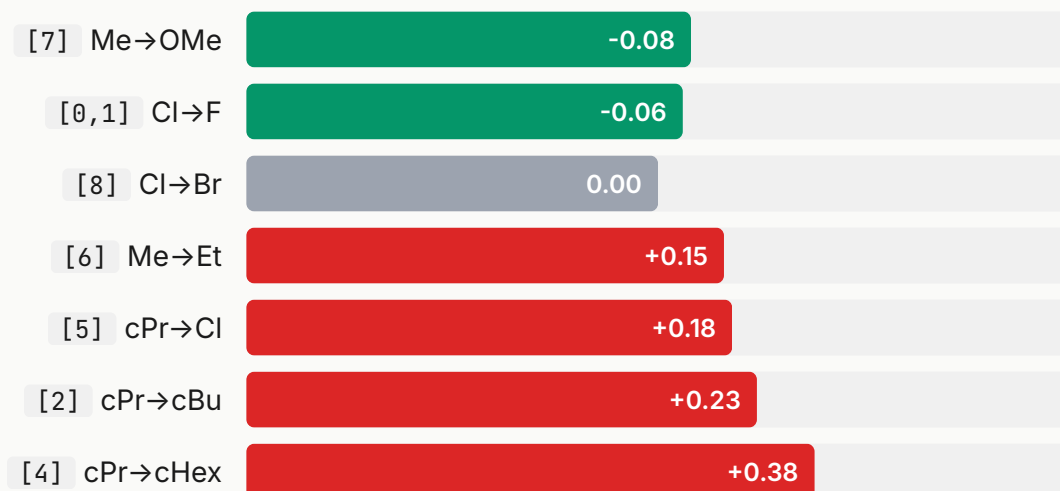
- Cherry-pick the 2–3 transforms with the best population means
- Stop after finding "good enough" improvements
- Never test transforms with neutral or positive population means

- Produce excellent mechanistic reasoning for what they do test

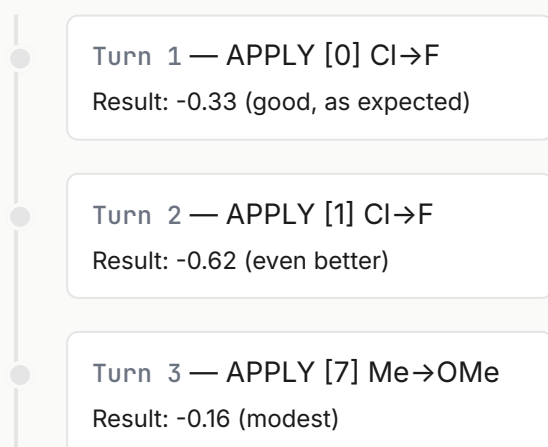
4. Task 001 Walkthrough: What the Model Sees vs. Misses

9 transforms available. The model must optimize intrinsic clearance (lower is better).

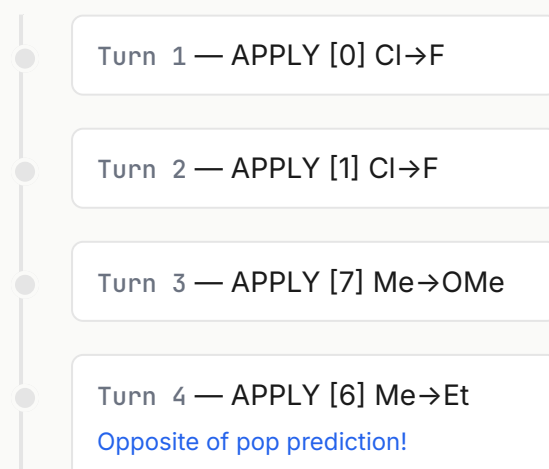
What the population statistics show:



Baseline (no nudge)



With nudge





Turn 4 — **DONE**

Skipped 6 transforms including the outlier

Coverage: 3/9 (33%)

Best found: -0.62



Turn 5 — **APPLY [8] Cl→Br**

OUTLIER FOUND! $\Delta = -1.94$ (5.2σ)



Turns 6-9 — Tests all remaining

Discovers cPr→cHex also gives -1.94

Coverage: 9/9 (100%)

Best found: -1.94

5. The Prompt Ablation: One Sentence Changes Everything

The nudge (added to system prompt):

"Population statistics reflect average behavior across many different molecules. Individual molecules can deviate dramatically from these averages due to specific structural context. A transform that looks neutral or unfavorable on average may be transformative for YOUR specific molecule. Do not skip transforms just because their population statistics look unpromising — test broadly before concluding."

TASK	BASELINE (SONNET)	WITH NUDGE (SONNET)
001 — 9 transforms	3/9 tested — FAIL	9/9 tested — PASS
009 — 6 transforms	2/6 tested — FAIL	6/6 tested — PASS

What this means

- **The failure is prompt sensitivity, not capability.** The model has excellent chemical reasoning — it just needs "permission" to explore.
- **Cascade effect:** Once nudged, encountering even one deviation from population predictions catalyzes exploration of all remaining transforms.
- **Better science:** The nudged model discovered richer SAR (e.g., cPr < cBu < cPent < cHex clearance trend) — invisible to the satisficing model.

6. Non-Additive SAR: The Underlying Phenomenon

We searched the full MMP database for transforms that behave as extreme outliers on specific molecules despite being well-characterized in the population.

2,235

Non-additive cliffs
($z \geq 3$, $n \geq 10$)

14

ADME endpoints
covered

0.65

Mean prediction error
(log units)

How population statistics mask non-additivity

The same transforms look completely normal in the population but behave as extreme outliers on specific molecules:

+0.66

Avg z-score in population
(looks normal)

-1.60

Avg z-score on cliff molecules
(extreme outlier)

7. SAR Epistasis: When Modifications Interact

Beyond single-transform cliffs, we found **36 synergistic quartets** across 5 ADME endpoints: cases where two modifications each worsen a property alone but improve it together. The chemical equivalent of genetic epistasis.

Parent (A) baseline	Change 1 only (B) +0.3 worse
Change 2 only (C) +0.2 worse	Both changes (D) -0.5 better!

Additive prediction: +0.5 (worse). Actual: -0.5 (better). Free-Wilson model fails completely.

Future task variant: "propose the combination"

Give the model both individual results (each worsening the property).
Ask whether it proposes trying both together anyway. Tests whether the model reasons about non-additive SAR or assumes Free-Wilson additivity.

8. The Broader Benchmark: Task Taxonomy

The exploration tasks are one tier of a larger benchmark testing medicinal chemistry reasoning.

TIER	TASK TYPE	WHAT IT TESTS	~N
1. Single-step Reasoning	property_delta	Predict property change + explain	350
	series_completion	Interpolate within congeneric series	210
	transform_ranking	Rank modifications by improvement	140
	tradeoff_analysis	Multi-endpoint effects	50
	transform_explain	Explain WHY a transform works	240
2. Strategic Multi-step	sacrifice_detection	Why was a worse intermediate accepted?	60
	strategic_planning	Choose best 2-step path (step 2 hidden)	60
	multi_objective_path	Navigate CLint/CYP tradeoffs	80
3. Interactive Exploration	exploration	Persist through non-additive SAR	22

9. Next Steps

Immediate

- Run all 11 well-hidden exploration tasks across Sonnet, Opus, GPT-4o, Gemini
- Systematic prompt ablation: vary nudge strength from none → subtle → explicit
- Run static benchmark (Tiers 1–2) through multi-model comparison

Research questions

- **Agentic frameworks:** Does wrapping the model in a structured exploration protocol (e.g., "always test N transforms before concluding") improve discovery without prompt nudging?
- **Epistasis quartets:** Can models learn to propose combinations of individually-failing modifications? (36 examples across 5 endpoints)
- **Fine-tuning signal:** Could training on exploration tasks with non-additive rewards improve the model's natural curiosity?
- **Cross-domain transfer:** Does the satisficing pattern hold for non-chemistry optimization tasks with deceptive landscapes?

For the paper

- Expert evaluation of static tasks (HTML report with chemist ratings)
- Auto-assessment vs. expert agreement correlation
- Representation perturbation (randomized SMILES) as memorization control

